

Friend or fold?

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1 Setting

We have a Kolmogorov dynamical system

$$\dot{x}_i = x_i(r_i + g_i(x, \alpha)), \quad (1)$$

where $g_i(x, \alpha)$ is a function for which the parameter α controls the strength of HOIs. The Jacobian at a feasible equilibrium $x^* > 0$ is

$$J(x^*) = D(x^*) M(x^*, \alpha), \quad (2)$$

where $D(x^*) = \text{diag}(x^*)$ and $M(x^*, \alpha)$ is the Jacobian of $g(x, \alpha)$ evaluated at x^* .

Further, we assume that $x^* = \mathbf{1}$ is a locally stable equilibrium of the system for each $\alpha \in [0, \alpha_{\max})$. We then imagine modifying the parameters $r(\eta)$ such that the equilibrium moves to

$$x(\eta) = \mathbf{1} + \eta d, \quad \|d\| = 1, \quad \eta \geq 0. \quad (3)$$

Since an interior equilibrium requires $r_i + g_i(x^*, \alpha) = 0$ for every i , this prescription determines the parameters uniquely, $r(\eta) = -g(x(\eta), \alpha)$; along the ray the equilibrium therefore always exists, and what can fail is either its positivity or its stability. As we step away from $x = \mathbf{1}$, one of two things can happen: (a) $x(\eta)$ leaves the positive cone (transcritical), or (b) $x(\eta) > 0$ loses stability.

First, let us analyze the case in which $x(\eta)$ touches the boundary of the positive cone. This happens whenever there exists a $d_i < 0$, at $\eta_i = -1/d_i$. Call $\eta_{\text{feas}} = 1/\max_i(-d_i)$ if any $d_i < 0$, and $\eta_{\text{feas}} = \infty$ otherwise; for d drawn uniformly on the sphere the latter case occurs with probability 2^{-n} , which is negligible for large n . Whenever $\eta \geq \eta_{\text{feas}}$ the equilibrium has left the positive cone. If stability was maintained up to this point, we have found a *transcritical bifurcation*: at $\eta = \eta_{\text{feas}}$ the interior branch meets the branch lying on the face $x_i = 0$ and the two exchange stability.

The loss of stability can further be separated into two main classes: (i) a Hopf bifurcation, in which a complex-conjugate pair of eigenvalues of J crosses the imaginary axis into the right half-plane without making J singular; (ii) a zero-eigenvalue bifurcation, in which there exists a critical η such that $\det J(x(\eta)) = 0$. The fold (codimension one) and its degenerations—cusp, Bogdanov–Takens, zero-Hopf—of higher codimension belong to this class. Since $\det J = (\prod_i x_i) \det M$ and $\prod_i x_i > 0$ in the interior, class (ii) occurs at an interior point if and only if $\det M(x(\eta), \alpha) = 0$.

For a fixed direction d and fixed α , the displacement η constitutes a single parameter. Generically, a one-parameter family encounters only codimension-one bifurcations, and the complete

list of codimension-one *local* bifurcations available to a positive equilibrium of a Kolmogorov system is: boundary transcritical ($x_i \rightarrow 0$), fold ($\det M = 0$), and Hopf. Tracking these three therefore suffices to locate the loss of stable feasibility,

$$\eta^*(d, \alpha) = \min(\eta_{\text{feas}}, \eta_{\text{det}}, \eta_{\text{Hopf}}), \quad (4)$$

where $\eta_{\text{feas}} = 1/\max_i(-d_i)$. Codimension-two points require a second defining condition and are therefore not encountered along a generic ray; they appear in the (η, α) plane as isolated organizing centers at which the corresponding codimension-one curves meet. We emphasize that this classification is local, in the sense that it exhausts the bifurcations detectable from $\text{spec } J(x(\eta))$; global codimension-one events (homoclinic orbits, saddle-node bifurcations of limit cycles, boundary crises) are equally generic in one parameter but leave the spectrum of the Jacobian unchanged, and may govern the fate of the system beyond a subcritical Hopf bifurcation.

In what follows we determine how α , the strength of HOIs, influences the values of η_{feas} and η_{det} .

2 A simplified model

Consider the simplified model

$$g_i(x, \alpha) = \sum_j A_{ij} x_j + \alpha \sum_j \sum_k B_{ijk} x_j x_k, \quad (5)$$

with $r = -A\mathbf{1}$, $B_{ijk} = B_{ikj}$ (the slices $B_{i..}$ are symmetric, which is without loss of generality, since only the part of B symmetric in (j, k) contributes to the quadratic form) and, in addition, the restriction

$$\sum_k B_{ijk} = 0 \quad \text{for all } i, j. \quad (6)$$

Note that $x^* = \mathbf{1}$ is then an equilibrium, since $r_i + g_i(\mathbf{1}, \alpha) = \alpha \sum_j (\sum_k B_{ijk}) = 0$. Writing

$$\mathcal{B}(x)_{ij} = \sum_k B_{ijk} x_k, \quad (7)$$

differentiation of (5) gives $M(x, \alpha) = A + 2\alpha \mathcal{B}(x)$, where the factor 2 follows from the slice symmetry of B . Constraint (6) states precisely that $\mathcal{B}(\mathbf{1}) = 0$, so that

$$M(\mathbf{1}, \alpha) = A, \quad J(\mathbf{1}) = D(\mathbf{1}) M(\mathbf{1}, \alpha) = A, \quad (8)$$

and $x^* = \mathbf{1}$ is locally stable for every $\alpha \geq 0$ if and only if A is Hurwitz. In this contrived parameterization we have chosen B in such a way that at the equilibrium $x^* = \mathbf{1}$ it does not enter the Jacobian. For example, B could be

$$B_{1..} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}, \quad B_{2..} = \begin{pmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{pmatrix}, \quad (9)$$

or, for a three-dimensional system,

$$B_{1..} = \begin{pmatrix} -\frac{5}{6} & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & -\frac{2}{3} & \frac{1}{6} \\ \frac{1}{3} & \frac{1}{6} & -\frac{1}{2} \end{pmatrix}, \quad B_{2..} = \begin{pmatrix} -\frac{1}{4} & -\frac{1}{4} & \frac{1}{2} \\ -\frac{1}{4} & \frac{7}{12} & -\frac{1}{3} \\ \frac{1}{2} & -\frac{1}{3} & -\frac{1}{6} \end{pmatrix}, \quad B_{3..} = \begin{pmatrix} \frac{13}{12} & -\frac{1}{4} & -\frac{5}{6} \\ -\frac{1}{4} & \frac{1}{12} & \frac{1}{6} \\ -\frac{5}{6} & \frac{1}{6} & \frac{2}{3} \end{pmatrix}. \quad (10)$$

For a given direction d , η_{feas} is fixed and does not depend on α . We now determine how α influences η_{det} . For $\det J(x(\eta), \alpha) = 0$ with $x(\eta) > 0$ we need $\det M(x(\eta), \alpha) = 0$. Because $\mathcal{B}$ is linear in its argument and $\mathcal{B}(\mathbf{1}) = 0$,

$$\mathcal{B}(x(\eta)) = \mathcal{B}(\mathbf{1}) + \eta \mathcal{B}(d) = \eta \mathcal{B}(d). \quad (11)$$

Note also that slice symmetry together with (6) implies $\sum_j B_{ijk} = 0$ for all i, k , whence $\mathcal{B}(x)\mathbf{1} = 0$ for *every* x : the matrix $\mathcal{B}(d)$ is always singular, with $\mathbf{1}$ in its kernel.

We may now rewrite the equation for the determinant, using $\det A \neq 0$ to factor A out:

$$\begin{aligned} \det M(x(\eta), \alpha) &= 0 \\ \det(A + 2\alpha \mathcal{B}(x(\eta))) &= 0 \\ \det(A + 2\alpha \eta \mathcal{B}(d)) &= 0 \\ \det(A(I + 2\alpha \eta A^{-1} \mathcal{B}(d))) &= 0 \\ \det A \det(I + 2\alpha \eta A^{-1} \mathcal{B}(d)) &= 0 \\ \det A \det(I + 2\alpha \eta Q(d)) &= 0, \end{aligned} \quad (12)$$

where $Q(d) := A^{-1} \mathcal{B}(d)$. Given that $\det A \neq 0$ because A is Hurwitz, and that $\alpha, \eta > 0$, singularity of M requires $-1/(2\alpha\eta) \in \text{spec } Q(d)$. Since $Q(d)$ does **not** depend on α , but only on the direction d , this means that $Q(d)$ must possess a real, negative eigenvalue, say $-\rho$ with $\rho > 0$. If the matrix has several such eigenvalues, the first crossing along the ray corresponds to the one farthest from zero, i.e. to the largest ρ . Then, for fixed α ,

$$\eta_{\text{det}} = \frac{1}{2\alpha\rho} > 0, \quad (13)$$

and $\eta_{\text{det}} = +\infty$ when $Q(d)$ has no real negative eigenvalue. Provided the critical eigenvalue is simple, $\det M$ changes sign at (13) and the crossing is a genuine fold; a double real eigenvalue would make $\det M$ touch zero without a sign change, a degeneracy of codimension two that does not occur for generic d .

For each d we therefore have a **fixed** η_{feas} and an η_{det} that depends inversely on α . Hence η_{det} decreases monotonically with increasing α , and the probability that $\eta_{\text{det}} < \eta_{\text{feas}}$ is non-decreasing in α . More formally:

Proposition 1 (Monotonicity in HOI strength). *Fix A Hurwitz and B satisfying (6), and let d be drawn from any distribution on S^{n-1} . Define*

$$\rho(d) = \max\{-\mu : \mu \in \text{spec } Q(d) \cap \mathbb{R}_{<0}\}, \quad Q(d) = A^{-1} \mathcal{B}(d), \quad (14)$$

with $\rho(d) = 0$ if $Q(d)$ has no real negative eigenvalue, and set $W(d) = \rho(d) \eta_{\text{feas}}(d)$, with the conventions $W(d) = 0$ whenever $\rho(d) = 0$ and $W(d) = +\infty$ whenever $\rho(d) > 0$ and $\eta_{\text{feas}}(d) = \infty$. Then for every $\alpha > 0$

$$\Pr[\eta_{\text{det}} < \eta_{\text{feas}}] = \Pr[W(d) > \frac{1}{2\alpha}], \quad (15)$$

which is non-decreasing in α , and is strictly increasing at α whenever the law of W assigns positive mass to every neighbourhood of $1/(2\alpha)$. Moreover

$$\lim_{\alpha \rightarrow \infty} \Pr[\eta_{\text{det}} < \eta_{\text{feas}}] = \Pr[\rho(d) > 0], \quad (16)$$

so that the probability saturates at the α -independent value $\Pr[\rho(d) > 0]$, which is in general strictly smaller than one.

Proof. Since $\mathcal{B}$ is linear and $\mathcal{B}(\mathbf{1}) = 0$, we have $M(x(\eta), \alpha) = A + 2\alpha\eta\mathcal{B}(d)$, so $\det M = 0$ requires $-1/(2\alpha\eta) \in \text{spec } Q(d)$, and therefore $\eta_{\det} = 1/(2\alpha\rho(d))$ with ρ independent of α . As η_{feas} is also independent of α , the event $\{\eta_{\det} < \eta_{\text{feas}}\}$ coincides with $\{W(d) > 1/(2\alpha)\}$. The threshold $1/(2\alpha)$ is strictly decreasing in α , so the sets $S_\alpha = \{d : W(d) > 1/(2\alpha)\}$ satisfy $S_{\alpha_1} \subseteq S_{\alpha_2}$ for $\alpha_1 < \alpha_2$; monotonicity of the probability follows, and strictness under the stated mass condition. Finally $\bigcup_{\alpha>0} S_\alpha = \{W > 0\} = \{\rho > 0\}$, and continuity of the measure from below along the nested union gives the limit. $\square$

Remark. The nesting $S_{\alpha_1} \subseteq S_{\alpha_2}$ holds pointwise in d , so the conclusion is distribution-free: no genericity or symmetry assumption on the law of d is required. Note also that $Q(d)\mathbf{1} = 0$ for every d , so at most $n - 1$ eigenvalues of $Q(d)$ are candidates for a fold, and whether a fold exists at all in a given direction is an α -independent property of $A^{-1}\mathcal{B}(d)$: the HOI strength controls only *where* along the ray the fold sits.

Remark. The case $n = 2$ is degenerate and should be read as an illustration rather than as evidence. There, (6) forces every slice to be proportional to $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, so that $\mathcal{B}(d) = (d_1 - d_2)ce^\top$ with $e = (1, -1)^\top$ and $c = (c_1, c_2)^\top$ the slice coefficients. Hence $Q(d)$ has rank one, its only nonzero eigenvalue is $\text{tr } Q(d) = (d_1 - d_2)e^\top A^{-1}c$, and a fold exists precisely when this quantity is negative. As it is linear in d , and $Q(-d) = -Q(d)$, exactly one member of each antipodal pair $\{d, -d\}$ admits a fold, so that $\Pr[\rho(d) > 0] = 1/2$ for any sign-symmetric law on S^1 .

Remark. Finally, a caveat on interpretation. The comparison above concerns $\eta_{\det}$ against η_{feas} only. Writing $D_d = \text{diag}(d)$,

$$J(\eta, \alpha) = (I + \eta D_d)(A + 2\alpha\eta\mathcal{B}(d)) \quad (17)$$

depends on η and on the product $\alpha\eta$ separately. The fold locus $\det M = 0$ is a function of $\alpha\eta$ alone, which is what makes the scaling (13) exact; the Hopf locus is not, and η_{Hopf} is finite already at $\alpha = 0$ whenever A fails to be D -stable. Consequently the Proposition does not imply that the probability of the fold being the *realized mechanism*, $\Pr[\eta_{\det} < \min(\eta_{\text{feas}}, \eta_{\text{Hopf}})]$, is monotone in α .

3 A more complex case